Software Engineering for CS

Week 5

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Switch off mobile phones
during lectures,
or
put them into silent mode

Next Assignment Deadline: 2-Nov-2023

Questions for Interview

At least 10 questions for each role

Questions must be drafted carefully by keeping the objective and outcome in mind

Explore similar projects

Meet the client

For products, meet the potential users

Submission includes

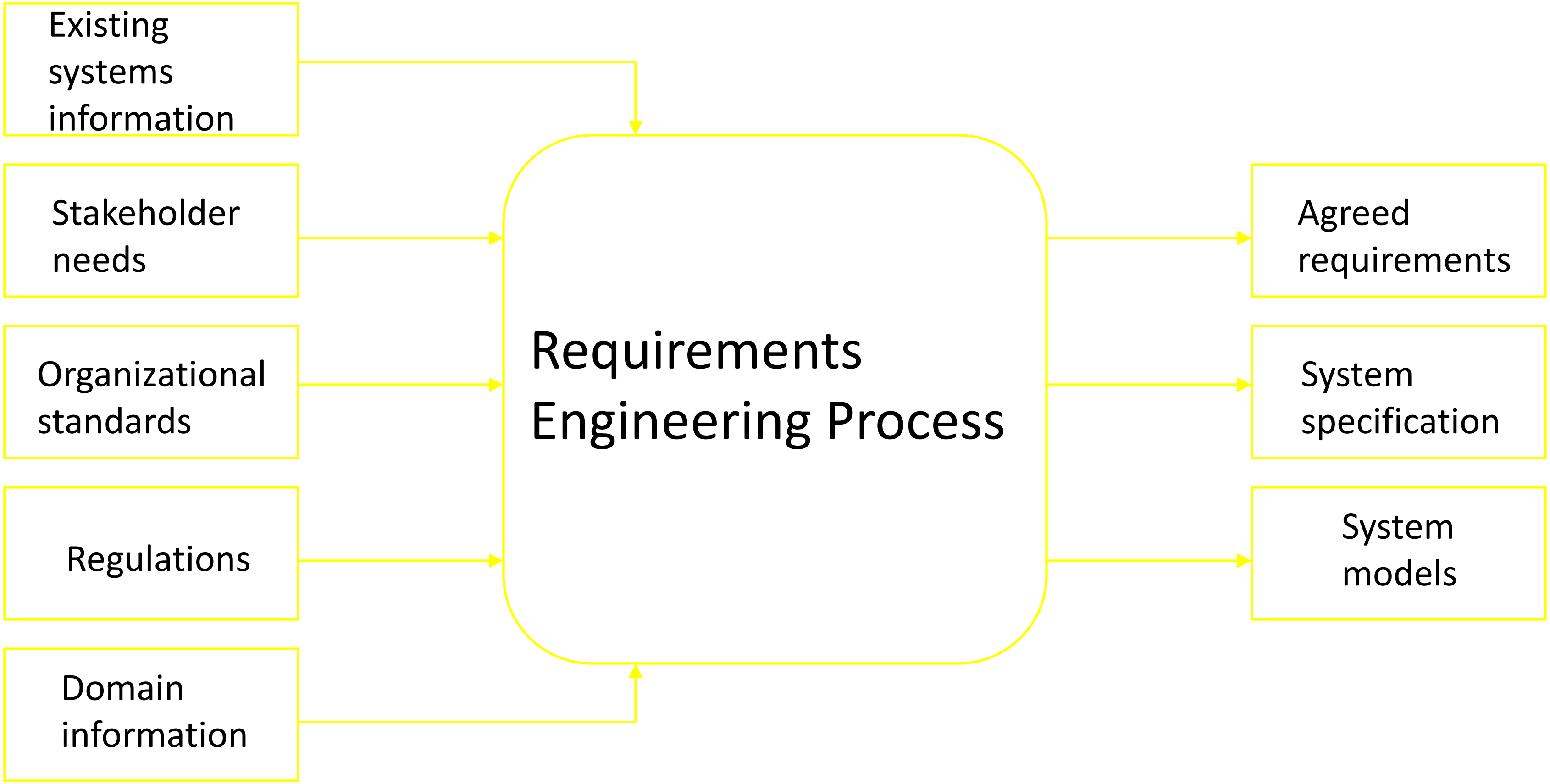
Set of questions for each type of user

Initial Requirements

Requirements Engineering Process

The process(es) involved in developing system requirements is collectively known as Requirements Engineering Process

RE Process - Inputs and Outputs



RE Process – Inputs

It includes:

Existing system information

- Information about the functionality of systems to be replaced
- Information about other systems, which interact with the system being specified

RE Process – Inputs

Stakeholder needs

- Description of what system stakeholders need from the system to support their work

Organizational standards

- Standards used in an organization regarding system development practice, quality management, etc.

RE Process – Inputs

Regulations

- External regulations such as health and safety regulations, which apply to the system

Domain information

- General information about the application domain of the system

RE Process – Outputs

It includes

Agreed requirements

- A description of the system requirements, which is understandable by stakeholders and which has been agreed by them

RE Process – Outputs

System specification

- This is a more detailed specification of the system, which may be produced in some cases

RE Process – Outputs

System models

- A set of models such as a data-flow model, an object model, a process model, etc., which describes the system from different perspectives

RE Process Variability

RE processes varies from one organization to another, and even within an organization in different projects

Unstructured process rely heavily on the experience of the people, while systematic processes are based on application of some analysis methodology , but they still require human judgment

Variability Factors - 1

There are four factors which count towards the variability of the Requirements Engineering Process

- Technical maturity
- Disciplinary involvement
- Organizational culture
- Application domain

Variability Factors - 2

Technical maturity

- The technologies and methods used for requirements engineering vary from one organization to other

Disciplinary involvement

- The types of engineering and managerial disciplines involved in requirements vary from one organization to another

Variability Factors - 3

Organizational culture

- The culture of an organization has important effect on all business and technical processes

Application domain

- Different types of application system need different types of requirements engineering process

RE Process - 1

Requirement Engineering Process has a formal starting and ending point in the overall software development life cycle.

Begins

- There is recognition that a problem exists and requires a solution
- A new software idea arises

Ends

- With a *complete* description of the external behavior of the software to be built

RE Process - 2

It is a continuous process in which the related activities are repeated until requirements are of acceptable quality

It is one of the most critical processes of system development

RE Process - 3

Based on the need of individual software projects and organizational needs, requirements engineering processes are tailored

An important point to remember is that

“There is no ideal requirements engineering process!”

Two Main Tasks of RE

There are two main tasks which needs to be performed in the requirements engineering process.

Problem analysis

- Analysis of a software problem

Product description

- Complete specification of the desired external behavior of the software system to be built. Also known as functional description, functional requirements, or specifications

Problem Analysis - 1

Problem analysis is the first and foremost task of requirements engineering process. It includes:

Brainstorming, interviewing, eliciting requirements

Identifying all possible constraints

Expansion of information

Problem Analysis - 2

Trading off constraints and organizing information

Complete understanding should be achieved

Product Description

Product description is another task of requirements engineering process. In this task we:

Make decisions to define the external behavior of the software product

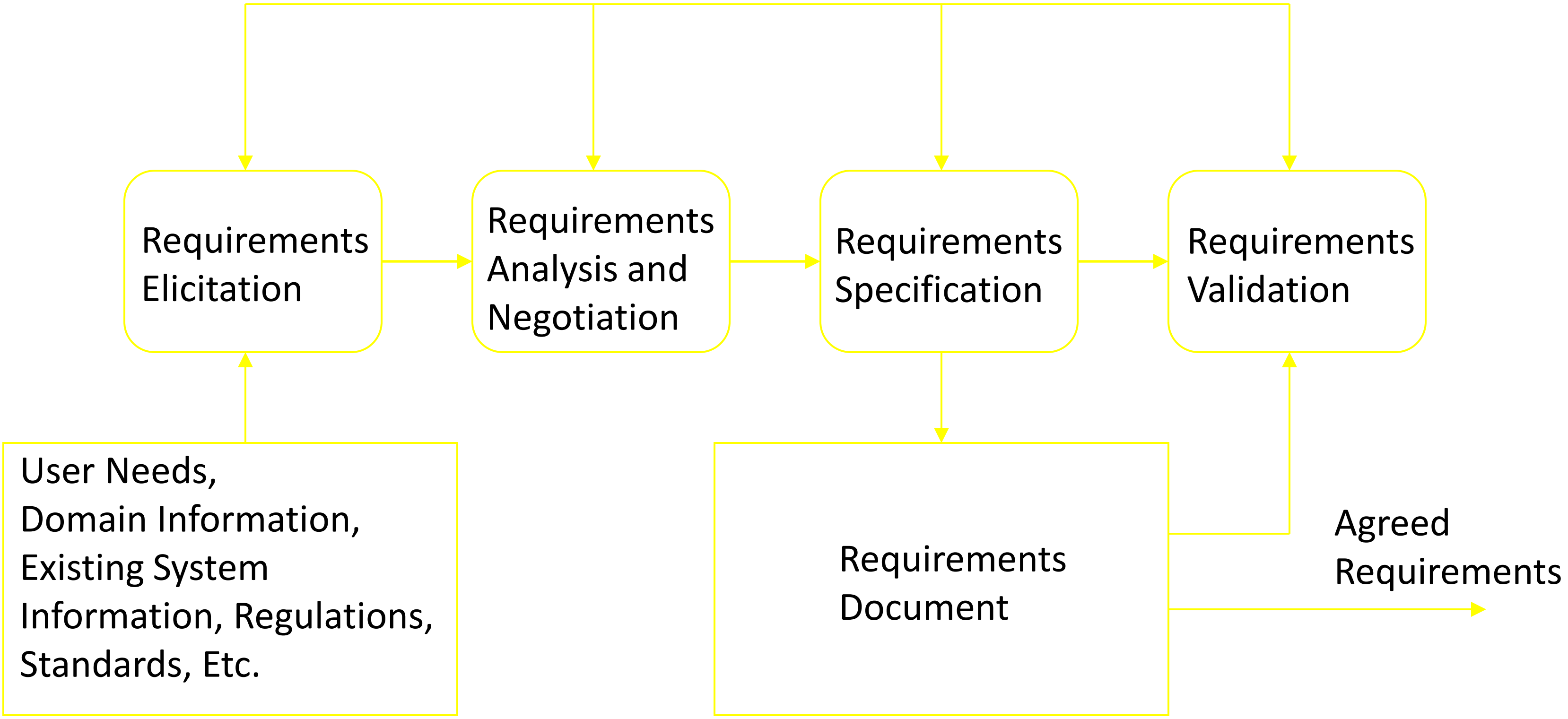
Organize ideas, resolve conflicting views, and eliminate inconsistencies and ambiguities

What Really Happens

It should be kept in mind that :

"Both problem analysis and product description run in parallel and iteratively throughout the requirements engineering process"

Requirements Engineering Activities



Requirements Elicitation

Requirements elicitation activity is performed by

Determining the system requirements through consultation with stakeholders, from system documents, domain knowledge, and market studies

Requirements acquisition or requirements discovery

Requirements Analysis and Negotiation - 1

Requirements analysis and negotiation activity is

- Understanding the relationships among various customer requirements and shaping those relationships to achieve a successful result
- Negotiations among different stakeholders and requirements engineers

Requirements Analysis and Negotiation - 2

Incomplete and inconsistent information needs to be tackled here

Some analysis and negotiation needs to be done on account of budgetary constraints

Requirements Specification

Requirements specification includes

Building a tangible model of requirements using natural language and diagrams

Building a representation of requirements that can be assessed for correctness, completeness, and consistency

Requirements Document

Detailed descriptions of the required software system in form of requirements is captured in the requirements document

Software designers, developers and testers are the primary users of the document

Requirements Validation

It involves reviewing the requirements model for consistency and completeness

This process is intended to detect problems in the requirements document, before they are used as a basis for the system development

Requirements Management

Although, it is not shown as a separate activity in RE Process, it is performed through out the requirements engineering activities.

Requirements management asks to identify, control and track requirements and the changes that will be made to them

Who are Actors?

Actors in a process are the people involved in the execution of that process

Actors are normally identified by their roles rather than individually, e.g., project manager, purchasing director, and system engineer

Actors in the RE Process - 1

Requirements engineering involves people who are primarily interested in the problem to be solved (end-users, etc) as well as people interested in the solution (system designers, etc.)

Another group of people, such as health & safety regulators, and maintenance engineers may be effected by the existence of the system

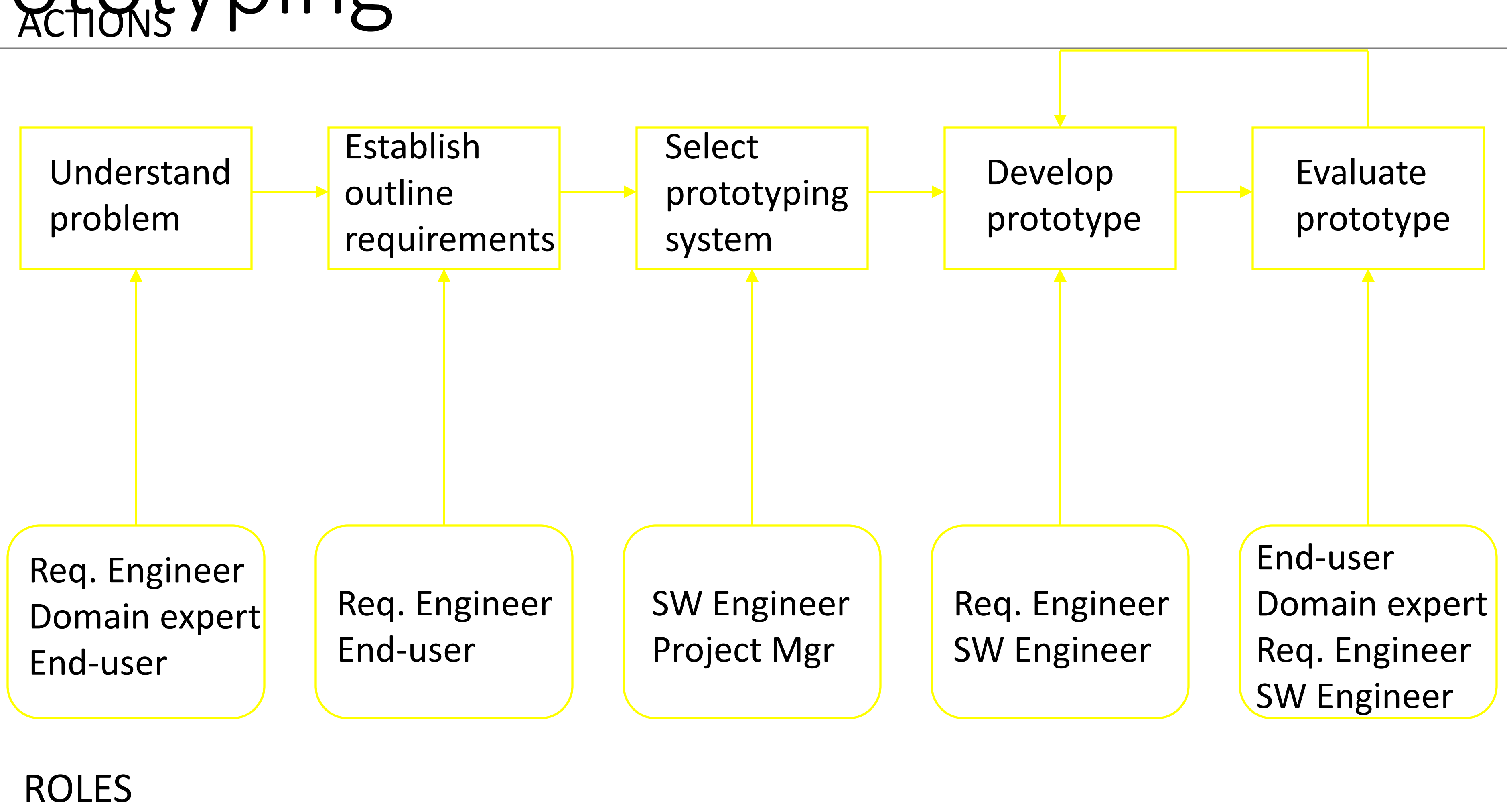
Actors in the RE Process - 2

Role-action diagrams are process models which show the actors associated with different process activities

They document the information needs of different people involved in the process

They use model of prototype software system as part of requirements elicitation process

Role-Action Diagram for Software Prototyping



Role Descriptions - 1

Role	Description
Domain Expert	Responsible for proving information about the application domain and the specific problem in that domain, which is to be solved

Role Descriptions - 2

Role	Description
System End-user	Responsible for using the system after delivery

Role Descriptions - 3

Role	Description
Requirements Engineer	Responsible for eliciting and specifying the system requirements

Role Descriptions - 4

Role	Description
Software Engineer	Responsible for developing the prototype software system

Role Descriptions - 5

Role	Description
Project Manager	Responsible for planning and estimating the prototyping project

Human and Social Factors

Requirements engineering processes are dominated by human, social and organizational factors because they always involve a range of stakeholders from different backgrounds and with different individual and organizational goals

System stakeholders may come from a range of technical and non-technical background and from different disciplines

Stakeholder Types

Software engineers

System end-users

Managers of system end-users

External regulators

Domain experts

Factors Influencing Requirements

Personality and status of stakeholders

The personal goals of individuals within an organization

The degree of political influence of stakeholders within an organization

Thanks